

Numerical Methods for ODEs: Exercise Week 4

Problem 1:

Implement the explicit Euler method in Python to solve the initial value problem

$$y'(t) = 3y(t) - y^2(t), \quad y(0) = y_0 \quad (1)$$

numerically on a time interval $0 \leq t \leq T$. Use a constant step size $t_{n+1} - t_n = h$.

Problem 2:

Confirm that

$$y(t) = \frac{3e^{3t}}{c + e^{3t}} \quad (2)$$

solves the differential equation. Determine the constant c depending on the initial value y_0 . Is this solution unique?

Problem 3:

1. Plot the global error of the explicit Euler method over time. Compare it against the theoretical bound provided by Theorem 27 and Remark 28 in the script.
2. Solve the initial value problem n times for n different constant time step sizes $h_0 > h_1 > \dots > h_n$ (say $n = 5$, for example) and compute the maximum global error for each simulation. Plot the maximum global error against the time step size in a plot where both axes are scaled logarithmically. What do you observe? Explain your observations in relation to the theoretically derived bound for the global error of the explicit Euler method.